

The shadow-size studio

Teacher guide and worked answers

Describe how moving an opaque card changes its shadow size when the source and screen stay fixed.

40 minutes. Use Lesson.html for interactive teaching or Lesson.pptx for editable slides. Slides.pdf is the static alternative. Select one pupil response route and print the shared evidence it requires. Detailed answers below are for staff.

Scheme of work

[_passsb/inputs/Build SOW 2026-2027.xlsx](#) | BUILD Weekly - Spring | C37

Explore how the size of a shadow changes; record findings.

Direct match to the named outcome. The lesson is a 40-minute exemplar, with optional greater depth; no accreditation completion is claimed.

Prior learning

Everyday objects can be described by what we observe.

Use a labelled diagram or evidence card to explain a choice.

Success evidence

Name the changed distance and what stays fixed.

Use data to describe the shadow-size pattern.

Predict a suitable card position for a stated target.

Equipment

Small low-power LED source; white screen; rigid opaque card 4 cm high; ruler or tape; stands; printed model data.

Preparation

Fix source and screen 100 cm apart. Mark card positions 20, 40 and 50 cm from the source.

Keep card and screen upright and parallel, with source aimed at card centre. Use a small LED to reduce blurred edges.

Pre-check space and visibility. Exact model numbers are not expected from every real torch.

Practical care

Use a cool low-power LED below face height. No lasers, hot lamps or sunlight.

Secure the screen and stands; keep floor paths clear and lit.

Do not move the light source or screen during comparisons; an adult checks the setup.

Ready to teach alternative

Use the constructed ideal-model data and diagrams. They describe a point-like source; real measurements can differ because source size and measurement edges matter.

Teaching sequence

Slide	Stage	Minutes	Focus
1	Opening	0-2	Same card. Two very different shadows.
2	Retrieval	2-5	Keep the shadow idea clear
3	I do	5-9	I move the card near the source
4	I do	9-12	I move it towards the screen
5	I do	Within stage	I measure the same dimension
6	We do	12-16	Find the pattern in the model
7	We do	16-20	Meet the studio target
8	Hinge	20-23	Which prediction follows this setup?
9	You do	23-25	Choose your investigation route
10	You do	Within stage	Use the evidence
11	You do	25-35	Make your own science record
12	You do	Within stage	Check and improve
13	Review	35-38	Lundy loop: improve our account
14	Exit	38-40	Three quick checks

Times are a suggested 40-minute route. Slides marked Within stage share the preceding stage allocation. The optional HTML timer starts only when selected.

1 Opening Same card. Two very different shadows.

Invite a prediction. State the studio brief: choose a card position to make a shadow about 10 cm high. No artistic drawing ability is required.

2 Retrieval Keep the shadow idea clear

Answers: blocks light transmission; on the screen; no, it is a region receiving less light. If needed trace the source-card-screen relationship.

3 I do I move the card near the source

Think aloud: "I keep the source and screen fixed. I only move the card. I trace the edge rays to the screen: their gap is wide. The shadow is large even though the card has not grown." This diagram is an ideal small-source model.

4 I do I move it towards the screen

Think aloud: "I check the same 4 cm card is used. Its edge rays now reach closer together on the fixed screen. In this setup, farther from the source means a smaller shadow." Do not generalise without stating what is fixed.

5 I do I measure the same dimension

Shares I do time. Model zero at the bottom of the shadow, read top at 8 cm. This is a constructed front-view example. Real blurred edges require a consistent chosen edge and a note about uncertainty.

6 We do Find the pattern in the model

Every number is constructed ideal-model data, not a measurement made by a class. Pairs place the position cards on a 100 cm line drawn on paper and match the heights. Largest at 20 cm; size falls as source-card distance increases.

Reveal: The model shadow height decreases: 20 cm, 10 cm, then 8 cm.

7 We do Meet the studio target

Use the three supplied model cards only. Each pair chooses then tests its choice against the table. 40 cm matches the target; 20 cm gives too large a shadow, 50 cm too small. Real practical pupils should check their measured outcome rather than promise exact equality.

Reveal: 40 cm from the source gives a 10 cm shadow in this model.

8 Hinge Which prediction follows this setup?

Insist on the stated controls. If pupils say “nearer screen,” make them point to source and screen before restating direction.

- A. The shadow becomes larger. - The card blocks a wider spread of rays before they reach the fixed screen.
- B. The card itself becomes taller. - The physical card stays 4 cm high; only its position changes.
- C. The shadow must stay the same height. - The edge-ray positions at the screen change when the card moves.

9 You do Choose your investigation route

Give one route only, selected with the pupil. The shared evidence page supports every route. Practical option: test the marked card positions with source and screen fixed. Measure the same top-to-bottom edge convention. Keep actual readings separate from the supplied model. A paper ruler-line and cards give the complete alternative.

10 You do Use the evidence

Shares the 12-minute independent stage. Practical option: test the marked card positions with source and screen fixed. Measure the same top-to-bottom edge convention. Keep actual readings separate from the supplied model. A paper ruler-line and cards give the complete alternative.

11 You do Make your own science record

Allow ten minutes of independent response after the two-minute setup. Practical option: test the marked card positions with source and screen fixed. Measure the same top-to-bottom edge convention. Keep actual readings separate from the supplied model. A paper ruler-line and cards give the complete alternative.

12 You do Check and improve

Shares the independent stage. Ask the pupil to point to the clue supporting their answer; accept a spoken correction or changed choice.

13 Review Lundy loop: improve our account

Give a private or paired option. Adopt one scientifically sound pupil revision in the class explanation and name the change. No public ranking.

Reveal: Nearer the source, with the same card and fixed source and screen.

14 Exit Three quick checks

Collect brief individual answers. Secure core requires the main idea and a relevant reason. Revisit the relevant worked diagram next lesson if the explanation reverses cause and effect.

Reveal: See the teacher answer guide for the three checks.

Answers and assessment

Retrieval 1-3

1 blocks light transmission. 2 on the screen. 3 no; it is an area receiving less light.

We do W1-W2

W1 shadow height falls as source-card distance grows: 20 cm at distance 20, 10 at 40, 8 at 50. W2 40 cm from source; other two produce 20 cm and 8 cm shadows.

Hinge

First option correct, with source and screen fixed. The card keeps the same actual height.

S1-S5

S1 A, 20 cm from source. S2 8 cm. S3 larger. S4 any two: source position, screen position, card size/orientation, source used. S5 40 cm.

T1-T4

T1 20, 10, 8 cm. T2 e.g. moving from 20 to 50 cm reduces model shadow from 20 to 8 cm. T3 card 40 cm from source, same 4 cm card upright and same source/screen 100 cm apart. T4 a real source has size; edges may blur; measurement/positioning can differ.

X1-X4

X1 larger than the 40 cm-position shadow. X2 between 10 and 20 cm, exclusive. Optional ideal calculation gives about 13.3 cm but is not required. X3 only position changes. X4 no; source-to-screen distance also affects size, so the fixed-screen comparison no longer applies unchanged.

Exit E1-E3

E1 larger. E2 source or screen position, or same card height. E3 40 cm from source.

Optional practical record

Actual values vary. Label as measured observations and retain any discrepancy from the model; do not overwrite them with the ideal numbers.

R1-R2

Responses vary. Credit a science-related revision or relevant next question. A pass or private response is acceptable; preferences have no fixed right answer.

Teaching and assessment notes

40 minutes: opening 2; retrieval 3; I do 7; We do 8; hinge 3; You do 12; review 3; exit 2. Zero-minute slides share the stage time.

BUILD follows the user's Year 3 science anchor with age-respectful materials for secondary pupils. Routes are adjustable access choices, not fixed groups or diagnoses.

Every core task can run from the printed evidence. Do not present constructed teaching evidence as observations pupils made. A pupil's spoken reasoning carries the same scientific value as writing.

For teacher understanding only: ideal shadow height = card height $\times$ source-screen distance $\div$ source-card distance. Here $4 \times 100 \div 20 = 20$ cm; $\div 40 = 10$ cm; $\div 50 = 8$ cm. Core pupils do not need this formula.

Use the phrase "in this setup". Distance claims are ambiguous unless the fixed source and screen and the moving card are named.

HTML We do: pupils choose a prediction, test it against the user-controlled SVG model or supplied evidence, then explain or revise. The shared paper cases and W1/W2 prompts provide the equivalent classroom discussion without a device. Any animation is a teaching model, not filmed experimental evidence.

Access and responsive teaching

Supported

Choose one route: short choices, word bank and pointing or adult-scribed explanation; keep the same science goal.

Standard

Use the evidence provided to record a decision and explain the science in short sentences.

Stretch

Apply the same core idea to a changed setup, identify a limitation and justify a better decision.

Regulation

Preview the sequence and equipment. Offer seated observation, a quiet paper route, partner rehearsal and a pass. Never require public performance or a darkened room.

Misconceptions to check

The card grows when the shadow grows.

The same card is used; its position changes how much of the spreading light is blocked.

Check: Ask which object was physically changed.

Nearer always means a smaller shadow.

With source and screen fixed, moving the card nearer the source makes a larger shadow in this setup.

Check: Name what nearer refers to.

Every torch must give the exact model numbers.

The ideal model simplifies source size and sharp edges; real observations can differ.

Check: Compare sample model data with an actual measurement label.

Word	Meaning
distance	The gap between two positions.
shadow height	The top-to-bottom size of the shadow.
model	A simplified account used to explain or predict.

Scientific references

Department for Education: science programmes of study

Checked 8 September 2026. Year 3 light outcomes anchor the BUILD content; ray explanations are carefully modelled as an accessible extension.